# Healthcare Machine Learning Project

Pima Diabetes · Heart Disease · Insurance Costs

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# Project Overview

### Goal

Apply the full supervised ML workflow to 3 real healthcare datasets — from raw data to evaluated and compared models.

### Tools

Python · scikit-learn · pandas · matplotlib · seaborn · GridSearchCV

### Tasks

2 Classification datasets
1 Regression dataset
2–3 models each

### Evaluation

Accuracy · F1 · ROC-AUC (classification)
MAE · RMSE · $R^2$ (regression)

*All hyperparameters tuned with GridSearchCV · 5-fold cross-validation throughout · Results compared to 3 published papers*

# The Three Datasets

**Classification**

## Pima Indians Diabetes

ROWS

768

FEATURES

8

TARGET

Diabetic / Not

*Hidden zeros replaced with column medians*

**Classification**

## Cleveland Heart Disease

ROWS

297

FEATURES

13

TARGET

Disease / No Disease

*Balanced classes*
*All features numeric*

**Regression**

## Medical Insurance Costs

ROWS

1,338

FEATURES

6

TARGET

Annual Charges ($)

*No missing values*
*Categorical encoding applied*

# Model Selection

| Dataset | Algorithm | Why |
|---|---|---|
| Diabetes | Logistic Regression | Simple interpretable baseline |
| Diabetes | Random Forest | Handles nonlinear feature interactions |
| Diabetes | MLP Neural Network | Complex pattern recognition |
| Heart Disease | KNN | Instance-based, balanced classes |
| Heart Disease | Decision Tree | Interpretable clinical rules |
| Heart Disease | Naive Bayes | Works well with independent features |
| Insurance | Linear Regression | Continuous target baseline |
| Insurance | Decision Tree Regressor | Captures nonlinear smoker effect |
| Insurance | Random Forest Regressor | Reduces overfitting vs single tree |

# Preprocessing & Hyperparameter Tuning

## Preprocessing

- Zero imputation — Diabetes dataset
  - Glucose, BMI, Insulin, BP & Skin Thickness zeros → column median
- StandardScaler applied for LR, KNN & MLP
- Label encoding (sex, smoker) + one-hot encoding (region)
- Stratified 80/20 train-test split (classification)
- 5-fold StratifiedKFold cross-validation

## Hyperparameter Tuning (GridSearchCV)

| Model | Key Parameters Searched |
|---|---|
| Logistic Reg. | C, solver |
| Random Forest | n_estimators, max_depth, max_features |
| MLP | hidden_layer_sizes, alpha, learning_rate |
| KNN | n_neighbors, weights, metric |
| Decision Tree | max_depth, criterion, min_samples |
| RF Regressor | n_estimators, max_depth, max_features |

# Results — Pima Indians Diabetes

*Binary Classification · 3 Models · After GridSearchCV Tuning*

| Model | Accuracy | Precision | Recall | F1 | ROC-AUC |
|---|---|---|---|---|---|
| Logistic Regression | 0.773 | 0.698 | 0.620 | 0.657 | 0.828 |
| Random Forest | 0.799 | 0.745 | 0.648 | 0.693 | 0.851 |
| MLP Neural Network | 0.786 | 0.729 | 0.630 | 0.676 | 0.839 |

**Key finding**

Random Forest was the top performer (AUC 0.851). Glucose and BMI are the two most important features by a wide margin, consistent with clinical expectation. Zero-imputation preprocessing measurably improved all three models vs raw data.

# Results — Heart Disease & Insurance Costs

## Cleveland Heart Disease

| Model | Acc. | F1 | AUC |
|---|---|---|---|
| KNN | 0.797 | 0.781 | 0.864 |
| Decision Tree | 0.780 | 0.767 | 0.780 |
| Naive Bayes | 0.814 | 0.790 | 0.876 |

*Naive Bayes outperforms — many clinical features are relatively independent of each other.*

## Medical Insurance Costs

| Model | MAE | RMSE | $R^2$ |
|---|---|---|---|
| Linear Reg. | $2,641 | $4,181 | 0.771 |
| Decision Tree | $2,895 | $4,730 | 0.738 |
| Random Forest | $2,427 | $3,745 | 0.863 |

*Random Forest $R^2 = 0.863$.*
*Smoker status is the dominant predictor by a large margin.*

**79.9%**

Best Accuracy
Diabetes (RF)

**81.4%**

Best Accuracy
Heart (Naive Bayes)

**0.863**

Best $R^2$
Insurance (RF)

**3/3**

Datasets won
by Random Forest

# Comparison to Prior Work

**Pima Diabetes**

*Kavakiotis et al. (2017) — Computational & Structural Biotechnology Journal*

Prior work: RF/SVM: 77–79% accuracy

**Our result: RF: 79.9% accuracy**

*Zero imputation likely contributed to slightly stronger results*

**Heart Disease**

*Detrano et al. (1989) — American Journal of Cardiology + Kaggle benchmarks*

Prior work: RF: 80–85% · KNN: 75–82%

**Our result: KNN: 79.7% · NB: 81.4%**

*Both within published ranges*

**Insurance**

*Rashid & Abdulrahman (2022) — Intl. Journal of Online & Biomedical Engineering*

Prior work: LR: $R^2$ 0.74–0.78 · RF: 0.85–0.87

**Our result: LR: $R^2$ 0.771 · RF: 0.863**

*Smoker dominance confirmed across both studies*

# Key Takeaways

### Random Forest won on all 3 datasets

Ensemble methods consistently outperformed linear and single-tree models across both classification and regression tasks.

### Preprocessing matters more than tuning

Zero-imputation on the diabetes dataset had a more noticeable effect on results than hyperparameter tuning alone.

### Results match published benchmarks

All three datasets produced results within or at the top of the ranges reported in the cited academic literature.

### Limitations to be aware of

Small sample sizes (Heart: 297 rows), population-specific data (Pima), and missing real-world features (Insurance) limit direct deployment.